

Transportation problem

Modified Distribution (MODI) method / u-v method

1. Solve the transportation problem using Modi method

	D ₁	D ₂	D ₃	Capacities
S ₁	5	1	8	14
S ₂	9	4	0	13
S ₃	17	6	7	30
Require-ments	31	11	15	

Step 1: checking for balanced rows & columns

$$\begin{aligned} 14 + 13 + 30 &= 57 \text{ (columns)} \\ 31 + 11 + 15 &= 57 \text{ (rows)} \end{aligned} \quad \left. \vphantom{\begin{aligned} 14 + 13 + 30 &= 57 \text{ (columns)} \\ 31 + 11 + 15 &= 57 \text{ (rows)} \end{aligned}} \right\}$$

Applying North West Corner method

	D ₁	D ₂	D ₃	Capacities
S ₁	5 (14)	1	8	14 → 0
S ₂	9 (13)	4	0	13 → 0
S ₃	17 (4)	6 (11)	7 (15)	30 → 26 0 → 15
Require-ments	31 → 17 4 → 0	11 → 0	15 → 0	57 57

no. of cells allocated = 5
 $m+n-1 = 4+1-1 = 5$

Both are same
 the solution is
 non degenerate

Initial feasible solution = $(14 \times 5) + (13 \times 9) + (4 \times 17)$
 $+ (11 \times 6) + (15 \times 7)$
 $= 70 + 117 + 68 + 66 + 105$
 $= 426$

checking for optimality using u-v method /
 Modi method.

$C_{ij} = u_i + v_j \rightarrow$ allocated cells.

5 (14)			$u_1 = 0$
9 (13)			$u_2 = ?$
17 (4)	6 (11)	7 (15)	$u_3 = ?$
$v_1 = ?$	$v_2 = ?$	$v_3 = ?$	

Allocated cells are $C_{11}, C_{21}, C_{31}, C_{32}, C_{33}$

$u_1 = 0$

$u_2 = 4$

$u_3 = 12$

$v_1 = 5$

$v_2 = -6$

$v_3 = -5$

Assume $u_1 = 0$

$C_{11} = u_1 + v_1$

$5 = 0 + v_1 \Rightarrow v_1 = 5$

$C_{21} = u_2 + v_1$

$9 = u_2 + 5 \Rightarrow u_2 = 4$

$C_{31} = u_3 + v_1$

$17 = u_3 + 5 \Rightarrow u_3 = 12$

$C_{32} = u_3 + v_2$

$6 = 12 + v_2 \Rightarrow v_2 = -6$

$C_{33} = u_3 + v_3$

$7 = 12 + v_3 \Rightarrow v_3 = -5$

Calculating opportunity cost of unallocated cells

3

	1	8
	4	0

$$\Delta_{ij} = C_{ij} - (u_i + v_j)$$

↳ for unallocated cells.

$$\begin{aligned}\Delta_{12} &= C_{12} - (u_1 + v_2) \\ &= 1 - (0 + -6) = +7\end{aligned}$$

$$\begin{aligned}\Delta_{13} &= C_{13} - (u_1 + v_3) \\ &= 8 - (0 + -5) = +13\end{aligned}$$

$$\Delta_{12}, \Delta_{13}, \Delta_{22} > \Delta_{23}$$

all have +ve values

the initial feasible

solution is optimal & unique.

$$\begin{aligned}\Delta_{22} &= C_{22} - (u_2 + v_2) \\ &= 4 - (4 + -6) = +6\end{aligned}$$

$$\begin{aligned}\Delta_{23} &= C_{23} - (u_2 + v_3) \\ &= 0 - (4 + -5) = +1\end{aligned}$$

Rules:

1. If Δ_{ij} has all positive values,
the solution is optimum and unique
2. If Δ_{ij} has a zero value,
the solution is optimum, not unique
3. If Δ_{ij} has negative values,
the solution is neither optimum, nor unique

2. Calculate optimum transportation cost using Modi method.

	D_1	D_2	D_3	Capacities
S_1	5	1	8	12
S_2	9	4	0	14
S_3	17	6	7	4
Requirements	9	10	11	

$$\left. \begin{aligned} \sum \text{Capacities} &= 12 + 14 + 4 = 30 \\ \sum \text{Requirements} &= 9 + 10 + 11 = 30 \end{aligned} \right\} \text{Balanced}$$

Solving
the above
using
least
cost
method

	D_1	D_2	D_3	Capacities
S_1	<u>5</u> ②	<u>1</u> ⑩	8	12 30
S_2	<u>9</u> ③	4	<u>0</u> ⑪	14 30
S_3	17 ④	6	7	4 0
Requirements	9 0	10 0	11 0	

$$\left. \begin{aligned} \text{Number of cells allocated} &= 5 \\ m + n - 1 &= 3 + 3 - 1 = 5 \end{aligned} \right\} \text{The solution is nondegenerate}$$

$$\begin{aligned} \text{Initial feasible solution} &= (2 \times 5) + (1 \times 10) + (9 \times 3) \\ &\quad + (0 \times 11) + (17 \times 4) \\ &= 10 + 10 + 27 + 0 + 68 \\ &= 115 \end{aligned}$$

Checking for optimality using u-v method

Allocated cells $\rightarrow C_{ij} = u_i + v_j$

5 ^②	1 ^⑩		$u_1 = 0$
9 ^③		0 ^⑪	$u_2 = ?$
17 ^④			$u_3 = ?$

$v_1 = ? \quad v_2 = ? \quad v_3 = ?$

$$u_1 = 0 \quad v_1 = 5$$

$$u_2 = 4 \quad v_2 = 1$$

$$u_3 = 12 \quad v_3 = -4$$

Let $u_1 = 0$

$$C_{11} = u_1 + v_1$$

$$5 = 0 + v_1 \Rightarrow v_1 = 5$$

$$C_{12} = u_1 + v_2$$

$$1 = 0 + v_2 \Rightarrow v_2 = 1$$

$$C_{21} = u_2 + v_1$$

$$9 = u_2 + 5 \Rightarrow u_2 = 4$$

$$C_{23} = u_2 + v_3$$

$$0 = 4 + v_3 \Rightarrow v_3 = -4$$

$$C_{31} = u_3 + v_1$$

$$17 = u_3 + 5 \Rightarrow u_3 = 12$$

checking opportunity cost

of unallocated cells $\rightarrow$

		8
	4	
	6	7

$$\Delta_{ij} = C_{ij} - (u_i + v_j)$$

$$\begin{aligned} \Delta_{13} &= C_{13} - (u_1 + v_3) \\ &= 8 - (0 + -4) = +12 \end{aligned}$$

$$\begin{aligned} \Delta_{22} &= C_{22} - (u_2 + v_2) \\ &= 4 - (4 + 1) = -1 \end{aligned}$$

$$\begin{aligned} \Delta_{32} &= C_{32} - (u_3 + v_2) \\ &= 6 - (12 + 1) = -7 \end{aligned}$$

$$\begin{aligned} \Delta_{33} &= C_{33} - (u_3 + v_3) \\ &= 7 - (12 + -4) = -1 \end{aligned}$$

There is one +ve and

three -ve values $\Delta_{22}, \Delta_{32}, \Delta_{33}$

Maximum negative number
is $\Delta_{32} = -7$

Putting together values obtained from allocated cells & unallocated cells,

5	1	8
9	4	0
17	6	7

From the above cells, the ^{maximum} negative values obtained is -7, from unallocated Δ_{32} lies in cell c_{32}

the $c_{32} = 6$

$\Delta_{32} \rightarrow c_{32} = 6$

Allocate +ve symbols to c_{32} & c_{11}
-ve symbols to c_{12} & c_{31}

5	1	8
9	4	0
17	6	7

In the cells with -ve symbols, we have 10 and 4

of the two, lesser value is 4

Select 4
 In -ve cells, subtract
 In +ve cells, add.

5	1	8
9	4	0
17	6	7

$4 - 4 = 0$

$10 - 4 = 6$

$2 + 4 = 6$

$0 + 4 = 4$

7

5	1	8	12
9	4	0	14
17	6	7	4
9	10	11	30 30

$$\begin{aligned}
 \text{Initial feasible solution} &= (6 \times 5) + (6 \times 1) + (3 \times 9) \\
 &\quad + (11 \times 0) + (4 \times 6) \\
 &= 30 + 6 + 27 + 0 + 24 \\
 &= 87
 \end{aligned}$$

$$\left. \begin{aligned}
 \text{No. of allocated cells} &= 5 \\
 m+n-1 &= 3+3-1=5
 \end{aligned} \right\} \text{The solution is nondegenerate}$$